

Consequences of academic selection for post-primary education in the United Kingdom: A systematic literature review

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Abstract

Academic selection for post-primary education is an important feature of the educational landscape in some parts of the United Kingdom, most notably in Northern Ireland and some areas of England, in addition to several other global contexts. This work systematically reviewed studies in a UK context which investigated the consequences of academic selection. The United Kingdom provided a suitable context for the review because both academic selection and all-ability post-primary education in comprehensive schools have featured, and continue to feature, in its education system, thus permitting a comparison of the relative merits of the two systems. Thirty-two publications met the inclusion criteria and were included in the narrative review. Results of the review indicate that, when relevant student and school characteristics are controlled for, academic selection has a negligible overall effect on academic achievement relative to all-ability comprehensive post-primary education. The findings also indicate that there is negligible difference between the capacity of the two systems to promote intergenerational social mobility, but that academic selection potentially reinforces and further entrenches socioeconomic disparities in educational outcomes. Possible gaps in the evidence base are highlighted and implications for further research, policy and practice are elucidated.

KEYWORDS

academic selection, consequences, grammar schools, United Kingdom

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Context and implications

Rationale for this study: This review was undertaken to create a robust, up-to-date synthesis of the evidence pertaining to the impact of academic selection in the United Kingdom.

Why the new findings matter: The findings are important because of ongoing debates about the future role of academic selection within the educational landscape of the United Kingdom.

Implications for policy makers and members of the public with an interest in education: The evidence reviewed suggests that there are benefits of a grammar school education for those who are fortunate enough to obtain one, particularly in relation to improved academic achievement and progression to higher education. However, it is also evident that there are some difficulties associated with academic selection in the UK context. The dominant issues of concern are the failure of academic selection to promote social mobility and the way in which it appears to reinforce and further entrench socioeconomic disparities in educational outcomes.

INTRODUCTION

Most education systems select students for higher education courses based on assessments of their academic ability since such assessments are deemed to represent a fair and meritocratic means of selecting candidates for relatively scarce places on competitive programmes (Kellaghan & Greaney, 2020). In many systems, post-primary students are also grouped for the teaching of either some or all subjects in different classes within the same school based on their prior academic attainment (Francis et al., 2017; Johnston & Wildy, 2018; Taylor et al., 2022). However, several jurisdictions, such as Northern Ireland and some parts of England, use a system of academic selection whereby students are selected by academic ability at a much younger age, around 10 or 11 years old, for post-primary education, so that students of different abilities attend different types of post-primary school.

During the 2023–2024 school year, there were 163 grammar schools in England, representing approximately 4.7% of all state-funded post-primary schools, and approximately 190,000 (5%) of state-funded post-primary students attended grammar schools (Department for Education, 2024). Furthermore, grammar schools are unevenly distributed in England, with significant concentrations in particular geographical areas, such as Kent, Lincolnshire and Buckinghamshire. Approximately one quarter of the local education authorities in England have at least one grammar school, while the remaining three quarters have no grammar schools. By contrast, grammar schools are ubiquitous in Northern Ireland, which has largely retained a selective education system. During the 2023–2024 school year in Northern Ireland, there were 66 grammar schools, representing approximately 34.4% of all state-funded post-primary schools, and these were attended by approximately 42.1% of all post-primary students (Department of Education in Northern Ireland, 2024).

In Northern Ireland and the relevant areas in England, students can opt to sit a high-stakes test of their academic attainment in the latter stages of primary education, known as a transfer test or 11+ test (henceforth referred to as the selection test), the results of which can determine the type of post-primary school they can attend. Those who perform sufficiently well in this test may have the option to attend grammar schools, which offer an academically focused education. Some evidence suggests that grammar school students are more likely to progress to higher education than their peers who attend other school types (Clark, 2010; Crawford, 2014; Lu, 2021). If more students obtain the required score in the grammar school

selection test than there are available places, grammar schools can apply additional criteria such as geographical proximity to the school. Students who do not sit the test, or who do not meet the grammar school admission criteria, attend non-selective schools, which offer a less academically oriented educational experience, with a greater emphasis on practical and vocational courses. Students cannot transfer from a non-selective school to a grammar school at any time solely by choice since grammar schools have strict admissions criteria, but transfers are possible if grammar school places are available, and students meet the admissions criteria. However, transfer from non-selective to selective schools in the UK context is less flexible than in some other education systems which allow more movement between school types, such as the USA (Cantley, 2024).

Academic selection for post-primary education in the United Kingdom has its roots in the 1944 Education Act for England and Wales, and the associated 1947 Education Act (Northern Ireland), which led to the provision of free post-primary education for all students (Cantley, 2024). Although grammar schools existed in England and Northern Ireland prior to these Acts, admission to them was generally on a fee-paying basis (Gardner, 2016). After the 1944 and 1947 Education Acts, a student could gain access to a free grammar school education based on a sufficiently meritorious performance in the selection test. Non-selective post-primary schools, referred to as secondary modern schools in England, were set up to cater for students who failed to gain, or did not seek, admission to a grammar school. Alongside grammar and secondary modern schools, a system of technical schools/colleges that focused on more vocationally relevant courses was inaugurated, but this was of limited success and was eventually phased out. As early as the 1950s, it became apparent that most grammar school students came from middle-class backgrounds, thus suggesting that social selection was a by-product of academic selection (Floud et al., 1956). Increasing criticism of academic selection during the 1960s and 1970s, led to its abolition in most of Great Britain, and the replacement of the bipartite secondary/grammar school system with all-ability comprehensive schools (Cantley, 2024). Despite these reforms, academic selection persists to the present day in Northern Ireland and some geographic areas of England. Advocates of academic selection argue that the use of selection tests to determine eligibility for a grammar school education represents a fair and objective method for allocating students to appropriate post-primary schools based on their talents and capabilities. More specifically, they posit that the academic selection process is unfettered by social background, and that it is an important mechanism for promoting social mobility (Cantley, 2024; Gallagher, 2021). In other words, it grants capable students from less privileged backgrounds access to greater educational and employment opportunities than may have been possible otherwise. However, the evidence in support of these claims, and the efficacy of selective education more generally, is disputed.

For example, in the UK context, some studies have found that students who attend grammar schools significantly outperform comparable non-selective school students in the General Certificate of Secondary Education (GCSE) examinations that students sit during the latter stages of compulsory schooling (Harris & Rose, 2013; Levačić & Marsh, 2007; Lu, 2020b; Shuttleworth & Daly, 2000). However, Gorard and Siddiqui (2018) reported that grammar school attendance confers no appreciable advantage in relation to GCSE outcomes when a range of school- and student-level attributes are controlled for. There are similar inconsistent findings in relation to the longer-term impact of academic selection, such as its effects on higher education participation and on social mobility. This is exemplified by the fact some researchers have reported that grammar school students are more likely to pursue higher education than those who attend non-selective schools (Clark, 2010; Gallagher & Smith, 2000), while others have disputed the existence of such a trend (Crawford, 2014; Lu, 2021). In a similar vein, some scholars have argued that academic selection promotes

social mobility (Mansfield, 2019), while others have challenged this viewpoint (Burgess et al., 2017; Buscha et al., 2021).

Although early academic selection, between the ages of 10 and 12 years, is not generally used in other Anglophone countries such as the USA, Canada, and Australia, it is a feature of the education systems of several European countries, such as Germany, Austria, and the Netherlands. In a similar vein to the UK context, individual studies of the consequences of selection in these jurisdictions have reported inconsistent findings. For example, in the German context, some studies have reported that early academic selection has positive effects on the learning progress of students attending the most selective schools, while others have concluded that early selection has a negligible impact on the learning progress of such students (Traini et al., 2021). However, the broad consensus in the international research indicates that, while ability grouping through academic selection has a negligible effect on overall student achievement at the system level relative to mixed ability teaching, academic selection increases the socioeconomic gradient in educational attainment (Bol & Van de Werfhorst, 2013; Horn, 2009; Terrin & Triventi, 2022; Van de Werfhorst & Mijs, 2010). Furthermore, international evidence suggests that early academic selection strengthens the correlation between socioeconomic background and students' probability of progressing to higher education (Parker et al., 2016) and it has a detrimental impact on the future earnings of low- and medium-ability students (Borghans et al., 2020).

Rationale for the review

Since academic selection and comprehensive education both feature in its post-primary educational landscape, the United Kingdom is a suitable context for research into the impact of academic selection relative to comprehensive education. However, the lack of consensus in the evidence base pertaining to the impact of academic selection in the United Kingdom suggests that a robust systematic review of the literature is warranted. Although there are some relevant existing evidence syntheses (Rix & Ingham, 2021; Terrin & Triventi, 2022), none of these synthesise quantitative evidence relating to academic selection consequences that is based on analyses of student-level educational outcomes in a UK context. For example, Rix and Ingham's (2021) systematic review of evidence relating to the impact of education selection based on notions of intelligence incorporated empirical evidence pertaining to all forms of selection, including both between-school and within-school grouping of students by ability. Furthermore, the included studies were not based exclusively on quantitative evidence derived from analyses of student-level educational outcomes in a UK context. Similarly, Terrin and Triventi's (2022) meta-analysis of a range of international evidence related to the consequences of all forms of ability grouping in schools, including both within-school and between-school grouping. The current systematic literature review is unique since it is based on a synthesis of relevant quantitative evidence derived from studies involving the analysis of individual student-level educational outcomes in the United Kingdom.

Academic achievement, as measured by performance in high-stakes public examinations, is an important determinant of students' future educational and vocational options (Kellaghan & Greaney, 2020). Since grammar schools select their students based on performance in a test of academic achievement, it is unsurprising that grammar school students generally tend to outperform those who have attended non-selective schools in later tests of academic achievement (Andrews et al., 2016; Gallagher & Smith, 2000). However, the use of value-added approaches to gauging the improvement in students' academic attainment, while controlling for a range of both student- and school-level attributes, including prior attainment, represents a fairer method for assessing the effect of academic selection on academic achievement. Given the academic focus of grammar

schools, it is important to establish if they lead to improved levels of academic achievement for their students without compromising the attainment of non-grammar students. Similarly, since academic achievement tends to influence higher education participation and earnings potential (Lu, 2021; Watts, 2020), it is logical to investigate the influence of academic selection on students' outcomes in these areas. Furthermore, given that the promotion of positive socio-emotional outcomes such as self-confidence, self-esteem and general emotional well-being is an important dimension of post-primary educational provision (Byrne et al., 2022; Tutt & Williams, 2021), an assessment of the impact of academic selection on students' outcomes in this area is warranted.

In many countries, students from more socioeconomically disadvantaged backgrounds achieve lower educational standards than their more privileged peers (Bruckauf, 2016; Liu et al., 2022). This is particularly problematic since the promotion of equity in educational outcomes between students from different socioeconomic groups is an important characteristic of high-performing education systems (Schleicher, 2018). Accordingly, the current review sought to examine the role of academic selection in promoting socioeconomic equity in students' educational outcomes. A commonly purported benefit of academic selection is that it promotes social mobility for capable students from less privileged backgrounds (Brown et al., 2021; Cantley, 2024; Gorard & Siddiqui, 2018; Harris & Rose, 2013). The veracity of this claim was critically evaluated in the current review of the evidence pertaining to the consequences of academic selection in the United Kingdom. For the review, social mobility was defined as 'assessing the chances of people climbing (or falling down) the income or social class ladder' (Major & Weiner, 2021, 301).

Therefore, the review focused on answering the following three research questions in the UK context:

1. What impact does academic selection have on educational outcomes such as academic achievement, progression to higher education, earnings, and socio-emotional outcomes?
2. To what extent does academic selection promote educational equity for students from different socioeconomic backgrounds?
3. Does academic selection promote social mobility?

MATERIALS AND METHODS

Overview

A systematic narrative review of existing literature on the consequences of academic selection in the United Kingdom was undertaken in accordance with the recommendations contained in the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021), and the PRISMA 2020 checklist for the review is included in Appendix 1. A strategy was established for the literature review, which included the formulation of an appropriate search string and the identification of relevant databases in consultation with the relevant Queen's University Belfast (QUB) subject librarian, together with the establishment of eligibility criteria for inclusion in the review.

Search string and databases

The following search string was used to search for scholarly literature, including both peer-reviewed and non-peer-reviewed reports as appropriate:

("academic selection" OR "grammar school**")
 AND
 (consequence* OR effect* OR outcome* OR impact* OR positive OR negative OR fail* OR success*)

AND
 (UK OR "United Kingdom" OR England OR NI OR "Northern Ireland" OR Scotland OR Wales)

The following databases, which were identified as being the most relevant by the QUB subject librarian, were searched on 12 April 2024 to locate scholarly literature using the above search string:

British Education Index
 Child Development & Adolescent Studies
 Education Source
 ERIC
 Proquest Dissertations & Theses
 PsycInfo
 Scopus
 Social Science Citation Index

Eligibility criteria

To be included in the narrative review, studies must have been written in English and published no earlier than 2000. Studies published prior to 2000 may not be relevant to the contemporary educational context due to curriculum changes, policy shifts and societal changes, and the inclusion of studies published since 2000 therefore ensures the review includes the most impactful and relevant research. Furthermore, included studies were required to have an explicit focus on the consequences of academic selection for post-primary education in the United Kingdom, that is, to focus on the consequences of students being selected by academic ability to attend different types of post-primary schools in the UK context. Since the review was concerned with the impact of educating students in different types of post-primary schools depending upon their ability levels, studies that focused solely on the consequences of allocating students to different classes within the same school based on ability level were excluded. Included studies also had to involve the analysis of individual-level quantitative data pertaining to student outcomes, and studies predicated on the analysis of aggregated data were excluded from the review. The rationale for the exclusion of the latter studies was to avoid drawing inaccurate inferences from the analysis of aggregated data due to variation in individual-level attributes that were not controlled for, that is, to avoid the ecological fallacy (Connolly, 2006). Qualitative studies were excluded because of the difficulty of synthesising qualitative and quantitative findings using a coherent analytical framework. To improve the robustness of the evidence synthesis, the following were excluded from the review:

1. Studies that focused on other individuals' (e.g., teachers' or parents') perspectives on student outcomes/consequences of academic selection.
2. Studies concerned with the validity/reliability of the academic selection procedure.
3. Policy/theoretical critiques of academic selection.

Finally, systematic reviews and meta-analyses were excluded from the review to avoid 'reviews of reviews'.

Therefore, the inclusion and exclusion criteria for eligible studies were as follows:

Inclusion criteria

1. Must be published no earlier than 2000.
2. Must be written in English.
3. Must have an explicit focus on the consequences of academic selection for post-primary education in the United Kingdom.
4. Must report on empirical quantitative research that entails the analysis of individual-level data pertaining to students' educational outcomes in the United Kingdom.

Exclusion criteria

1. Studies that focus solely on the consequences of allocating students to different classes within the same school based on ability.
2. Studies that do not entail the analysis of student-level quantitative data (e.g., those that entail analysis of aggregated data).
3. Studies that report on other individuals' (e.g., teachers' or parents') perspectives on student outcomes/consequences of academic selection.
4. Studies relating to the validity/reliability of the academic selection procedure.
5. Policy/theoretical critiques of academic selection.
6. Systematic literature reviews/meta-analyses.

Data extraction and analysis

The initial database searches returned 375 records, and these were augmented by 12 further reports obtained by a manual search, thus giving 387 records in total. EndNote 21 software was used to manage the records, including the removal of duplicates using the functionality provided within the software and most aspects of the screening process. Following the initial searches and the removal of duplicate records, one of the authors screened the remaining 253 reports by title and abstract. To reduce the probability of bias and error, a second author also screened a 20% random sample of the non-duplicate records by title and abstract. This process was performed by extracting the relevant details for the 20% random sample from EndNote 21 into a Microsoft Excel spreadsheet for independent screening by the second author. Inter-rater reliability was measured using Cohen's Kappa, and the calculated value of 0.916 demonstrated an excellent level of inter-rater reliability (Cicchetti, 1994). For those reports where the two authors disagreed, differences of opinion were discussed to arrive at a joint decision on the inclusion or exclusion of each relevant report. Based on title and abstract screening, 36 reports met the eligibility criteria.

Full-text screening of the 36 remaining reports was conducted by the author who screened all 253 reports by title and abstract, resulting in the inclusion of 32 studies in the final review. The following details were extracted from each included study:

1. Author(s).
2. Year of publication.
3. Number of individuals and data source.
4. Main outcome(s) investigated.
5. Analytical approach.
6. Summary of main findings.
7. Main limitations.
8. Trustworthiness rating.

The results were collated in a table and narrative analyses were performed in relation to:

1. The impact of selective education on educational outcomes such as academic achievement, progression to higher education, earnings, and socio-emotional outcomes.
2. The capacity of selective education to promote educational equity for students from different socioeconomic backgrounds.
3. The capacity of selective education to promote social mobility.

It was considered inappropriate to perform any type of quantitative synthesis on the included studies because they were undertaken for different purposes, and in a range of contexts, using different methodological approaches and data analysis methods. Furthermore, many of the included studies had weaknesses that reduced confidence in their reported conclusions and/or did not report all the necessary information to perform a quantitative synthesis. The trustworthiness of each included study was assessed on a scale of 0 to 4, with a higher rating indicating a greater level of trustworthiness. These ratings were obtained by assessing each study using the research design, scale, missing data, and measurement quality dimensions of Gorard's trustworthiness 'sieve' (Gorard, 2024). Each study was assessed by gauging the appropriateness of the research design for the research question(s) addressed, the number of cases in the smallest group used in the analyses, the proportion of missing data and its likely impact on findings, and the quality of the measurements used in the analyses. Initially, each study was assigned a rating for the appropriateness of its research design on the scale 0 to 4, where 0 indicates low quality and 4 indicates high quality. The study was then successively assessed on the same 0 to 4 scale in relation to the number of cases in the smallest comparison group, the proportion of missing data and its likely impact on findings, and measurement quality. If the assigned rating for any dimension exceeded the previous dimension rating, the composite rating was set equal to the previous rating, but equal to the dimension rating otherwise. Therefore, the final composite rating gave an overall estimation of the trustworthiness of the study on the 0 to 4 scale, where 0 denotes low level of trustworthiness and 4 denotes high level of trustworthiness.

The findings from the narrative review were critically evaluated and interpreted in relation to the insights they offered into the consequences of academic selection for students' educational outcomes in the United Kingdom.

RESULTS

Included studies

The selection process for the included studies is summarised in Figure 1, in line with PRISMA recommendations (Page et al., 2021), while Appendix 2 presents a summary of each included study, including author(s), year of publication, number of individuals and data source, main outcome(s) investigated, summary of main findings, main limitations, and trustworthiness rating. In Figure 1, the 375 records identified through database searches were obtained from the following databases:

1. British Education Index: $n=26$
2. Child Development & Adolescent Studies: $n=10$
3. Education Source: $n=88$
4. ERIC: $n=57$
5. Proquest Dissertations & Theses: $n=42$
6. PsycInfo: $n=16$

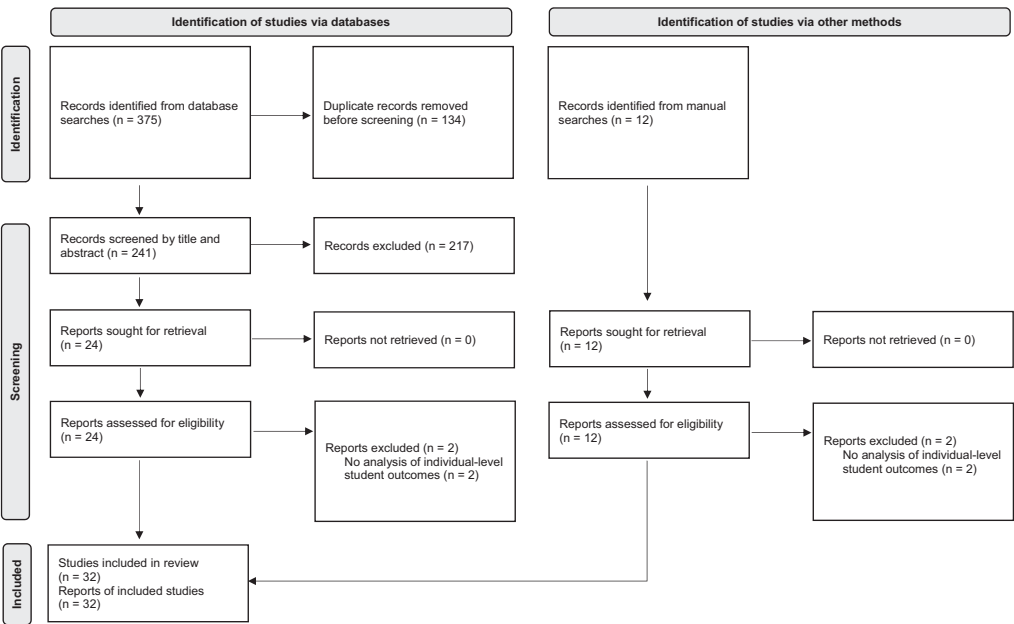


FIGURE 1 PRISMA diagram for selection of included studies.

7. Scopus: $n=89$

8. Social Science Citation Index: $n=47$

The majority of the 32 studies included in the review were large scale, with sample sizes ranging from 856 to 4,363,600. However, there was considerable variation in the quality of the studies, as reflected by the ratings that were assigned to them using Gorard's trustworthiness sieve (Gorard, 2024). No studies were given a rating of 4 and just 11 studies were assigned a rating of 3. A further 15 studies received a rating of 2, four studies were rated as 1, and the two remaining studies were given 0 ratings. The relatively small proportion of more robust studies, that is, those rated as 3 or 4, is noteworthy. This is mainly attributable to the relative dearth of strong research designs that permitted causal relationships to be established between academic selection and the reported findings.

A randomised controlled trial (RCT) is the gold standard for establishing causal relationships in educational research (Connolly et al., 2017). In the context of investigating the consequences of academic selection, an RCT would entail random allocation of students to grammar or non-grammar schools, and comparison of the relative change in students' outcomes over a given period. Obviously, it is not viable to conduct an RCT of this type in the context of academic selection, but there are alternative research designs that permit causal claims to be asserted, such as a regression discontinuity design (RDD) (Fan & Nowell, 2011) or propensity score matching approaches (Perry, 2017). In an RDD, borderline candidates who attain selection test scores close to the cut score are assigned to intervention or control groups depending upon whether they attain the cut score and are admitted to a grammar school (intervention group) or achieve a lower score and are denied admission to a grammar school (control group). Given the similarities between students in the two groups, an RDD minimises the probability of the existence of unmeasured baseline differences between the two groups and increases the likelihood that any observed differences between the groups is caused by academic selection. Propensity score matching, on the other hand, entails creating matched samples of comparable groups of students who attend grammar and

non-grammar schools, where the matching of the two groups is based on a range of variables that predict group membership. Again, any observed difference between the groups can be causally linked to academic selection with greater confidence.

In the absence of approaches such as RDD or propensity score matching, studies investigating the consequences of academic selection in the United Kingdom were mainly based on correlational analyses, which cannot permit causal relationships to be conclusively established. Such studies were assigned a maximum rating of 2 using Gorard's trustworthiness sieve. The main reasons for the assignment of trustworthiness ratings of 0 or 1 included non-consideration of relevant factors that could have impacted on the findings and/or significant deficiencies in the research design. However, the more robust included studies frequently had other limitations that compromised their trustworthiness, such as the use of other prior attainment scores as proxies for selection test scores in an RDD. Although there was a dearth of more trustworthy studies in the current review, that is, those with trustworthiness ratings of 3 or 4, it is important to note that the conclusions of studies with lower, non-zero trustworthiness ratings, that is, ratings of 1 or 2, broadly resonated with the findings reported by the higher quality studies and they therefore warrant consideration.

A summary of the main findings of the review in each of the three areas considered is presented below.

Impact of selective education on educational outcomes

Academic achievement

Just over half of the included studies (17) investigated the impact of academic selection on academic achievement. These studies generally reported that, when relevant student and school characteristics were controlled for, academic selection had a negligible overall effect on academic achievement relative to all-ability comprehensive post-primary education (Andrews et al., 2016; Atkinson et al., 2006; Clark, 2010; Galindo-Rueda & Vignoles, 2005; Gorard & Siddiqui, 2018; Lu, Anders, et al., 2024). Although the consensus view was that selection had little overall impact on achievement, most of the reviewed evidence concluded that grammar school attendance led to small/moderate positive effects on student achievement compared to the achievement of similar students who attended comprehensive schools (Andrews et al., 2016; Atkinson et al., 2006; Coe et al., 2008; Harris & Rose, 2013; Levačić & Marsh, 2007; Lu, 2020b; Lu & Siddiqui, 2024; Pastore & Jones, 2023; Schagen & Schagen, 2003; Sullivan et al., 2018). In addition, Sullivan et al. (2018) reported that grammar school students achieved significantly better academic qualifications than comparable comprehensive students both in the short term, at ages 16 and 18, and in the longer term, as reflected by the highest level of qualification achieved by age 42. However, many studies reported that the achievement-related benefits of attending a grammar school were counterbalanced by small negative achievement effects for students in non-grammar schools within selective areas relative to comparable students who attended comprehensive schools (Andrews et al., 2016; Atkinson et al., 2006; Harris & Rose, 2013; Levačić & Marsh, 2007; Sullivan et al., 2018). In Northern Ireland, where academic selection is ubiquitous, grammar students outperformed their non-grammar peers by a significant margin (Early, 2020; Shuttleworth & Daly, 2000).

There are some noteworthy nuances in the reviewed evidence pertaining to the impact of academic selection on achievement. Firstly, Lu and Siddiqui (2024) found that there was no systematic relationship between the magnitude of the positive grammar school achievement effect and the proportion of grammar school places within a particular geographical area. However, Andrews et al. (2016) reported that the positive grammar school effect was smallest for those areas with the highest proportion of available grammar school places,

which is logical given that there is likely to be less competition for grammar school places in such areas. Secondly, Lu, Anders, et al. (2024) concluded that academic selection disadvantaged high attaining students since such students were significantly less likely to obtain at least five GCSE grades A*–A (i.e., at least five GCSEs with one of the top two available grades) than comparable comprehensive students were. However, this conclusion contradicts Galindo-Rueda and Vignoles' (2005) finding that the highest ability students, especially girls, achieve significantly higher academic standards in grammar schools than comparable students in comprehensive schools. Nevertheless, the more contemporary focus of Lu, Anders et al.'s (2024) research means that greater credence can be given to their findings than those of Galindo-Rueda and Vignoles (2005), which pertained to individuals who were born in 1958 and who were educated in the 1960s and 1970s. Finally, but importantly, Coe et al. (2008) noted that, although grammar students made greater progress than non-grammar students did during their grammar school careers, they also made greater progress over the course of their primary school education. This points to the existence of differences between the two groups of students that have not been accounted for in the statistical models and thus reduces confidence in the existence of a positive grammar school effect.

Higher education participation and outcomes

A quarter of the included studies (8) examined the impact of academic selection on higher education participation/outcomes. Half of these studies reported that grammar school attendance increased the likelihood of students' participation in higher education relative to comparable students who attended other types of schools (Burgess et al., 2017; Clark, 2010; Crawford, 2014; Lu, 2021). Some of these studies also noted that grammar school students were significantly more likely to attend prestigious Russell group universities than comparable peers from other state school sectors (Burgess et al., 2017; Crawford, 2014; Lu, 2021). In less robust studies, which failed to control for relevant factors that may have impacted on their findings, Mansfield (2019) and Montacute and Cullinane (2018) reported similar conclusions regarding the increased probability of higher education participation for grammar students compared to those from other state schools. Capsada-Munsech and Boliver (2024), on the other hand, found that grammar students were no more likely to participate in higher education, or to attend prestigious Russell group universities, than students from non-selective state schools with comparable GCSE and A-level attainment. However, controlling for students' A-level attainment may be over-controlling given the strong correlation between A-level attainment and higher education participation. Interestingly, Lu, Dai et al. (2024) concluded academic selection strengthened the relationship between family background and higher education participation, increasing the probability of students from more privileged backgrounds progressing to higher education and correspondingly reducing the likelihood of higher education participation for those from less privileged backgrounds. Despite their relative advantage in the higher education participation stakes, grammar school students were found to be significantly less likely to finish their courses and to attain good degree classifications than comparable non-selective students (Burgess et al., 2017; Crawford, 2014).

Earnings

A relatively small number of the included studies (3) focused on the impact of academic selection on future earnings potential. The findings suggested that academic selection had limited impact on longer-term labour market and earnings-related outcomes (Birkelund et al., 2021; Pastore & Jones, 2023). Nevertheless, it is noteworthy that Burgess et al. (2020)

concluded that the adult earnings of individuals who grew up in selective education systems showed significantly greater inequality than for those who grew up under a comprehensive system.

Socio-emotional outcomes

The effect of academic selection on students' socio-emotional outcomes was also the focus of a comparatively small number of the included studies (4) (Ahmavaara & Houston, 2007; Gallagher & McKeown, 2000; Jerrim & Sims, 2019a, 2020). The two more robust studies concluded that there were minimal differences in socio-emotional outcomes such as well-being and self-esteem between grammar and comparable non-grammar students (Jerrim & Sims, 2019a, 2020). However, within selective areas of England, Jerrim and Sims (2019a) found grammar students to have significantly greater future educational aspirations than their non-grammar counterparts, which also resonates with the conclusions of Ahmavaara and Houston (2007). Furthermore, grammar students from Northern Ireland had significantly lower academic self-concepts than their comparable non-grammar peers (Jerrim & Sims, 2020), but non-grammar students from Northern Ireland perceived their schooling to be more relevant to their future careers and had higher perceived levels of social integration than their grammar peers (Gallagher & McKeown, 2000).

Capacity of selective education to promote educational equity for students from different socioeconomic backgrounds

Several of the included studies looked at the differential impact of academic selection on students from more and less privileged socioeconomic backgrounds (e.g., Andrews et al., 2016; Burgess et al., 2020; Coe et al., 2008; Early, 2020; Gorard & Siddiqui, 2018; Lu, 2020a; Lu, Dai, et al., 2024). In these studies, eligibility for free school meals (FSM) was generally used as a proxy for students' socioeconomic backgrounds since those from more socially disadvantaged backgrounds are more likely to be FSM eligible. Generally, the positive effect on academic achievement of attending a grammar rather than a non-grammar school was greater for FSM eligible than non-FSM eligible students, but the negative achievement effect of not attending a grammar school was also larger in magnitude for FSM eligible than non-FSM eligible students.

Given the under-representation of students from lower socioeconomic backgrounds in grammar schools (Andrews et al., 2016; Jerrim & Sims, 2019b; Levačić & Marsh, 2007; Lu, 2020c), the poverty attainment gap—that is, the difference between the attainment of FSM eligible and non-FSM eligible students—was greater in areas which retained academic selection than in other areas (Andrews et al., 2016; Gorard & Siddiqui, 2018). This suggests that the harmful effects experienced by FSM eligible students who failed to secure grammar school places in selective areas negated any benefits of grammar school attendance by FSM eligible students. Furthermore, academic selection strengthened the correlation between students' earlier and later academic attainment (Lu, Dai et al., 2024) and led to significantly greater inequality in the adult earnings of individuals who grew up in selective education systems compared to comprehensive systems (Burgess et al., 2020). Therefore, the evidence is clear: academic selection does little to promote educational equity for students from different socioeconomic backgrounds. In fact, it does the opposite since it serves to reinforce and further exacerbate socioeconomic disparities in educational outcomes.

Capacity of selective education to promote social mobility

Since the promotion of social mobility is one of the main purported benefits of academic selection (Cantley, 2024), it is important to critically evaluate the evidence pertaining to the social mobility consequences of academic selection. This is particularly pertinent given the under-representation of students from lower socioeconomic backgrounds in grammar schools (Andrews et al., 2016; Jerrim & Sims, 2019b; Levačić & Marsh, 2007; Lu, 2020c). After all, those from less privileged socioeconomic origins will be unable to reap the benefits associated with attending a grammar school without securing a grammar school place.

Mansfield (2019) used progression to prestigious higher education institutions as a proxy for the promotion of social mobility and claimed to offer evidence in support of the hypothesis that academic selection does indeed promote social mobility. However, Mansfield's claims are compromised by his failure to consider the outcomes for students who did not secure a grammar school place in selective areas and for students who did not progress to higher education. Just two of the included studies investigated the grammar school social mobility hypothesis in a more robust manner, using a more rigorous approach than relying on progression to higher education as an indicator of social mobility. Boliver and Swift (2011) found selective education to offer no overall statistically significant income or social class mobility advantage relative to comprehensive education for individuals from any socioeconomic background. In particular, the negative effects associated with attending a secondary modern school were found to offset any positive grammar school effect for low origin individuals. These findings resonate with Buscha et al.'s (2023) conclusion that the transition from a selective to a comprehensive education system in England had no statistically significant impact on intergenerational social mobility.

DISCUSSION

The negligible overall effect of academic selection on academic achievement compared to the comprehensive model of post-primary education resonates with the international evidence base on the impact of ability grouping on achievement. For example, in an extensive meta-analysis of the impact of ability grouping on students' academic achievement relative to mixed-ability teaching in different international contexts, Terrin and Triventi (2022) concluded that ability grouping had no statistically significant effect, at the 5% level, on students' overall or subject-specific achievement. However, they also found that between-school ability grouping through academic selection had a statistically significant positive effect (at the 0.1% level) on overall achievement relative to within-school ability grouping of the type that frequently occurs in comprehensive schools. Although this suggests that academic selection is more effective at increasing overall academic achievement than the within-school ability grouping arrangements that are typical in comprehensive contexts, the extremely small associated effect size (0.023) effectively means the difference is inconsequential.

The findings from the current review on longer-term educational outcomes, such as higher education participation and future earnings potential, are also consistent with the research evidence from other international contexts. The role of academic selection in strengthening the relationship between family background and higher education participation, with students from more privileged backgrounds being more likely to progress to higher education than those from less privileged backgrounds, aligns with Parker et al.'s (2016) conclusions about the link between socioeconomic background and higher education aspirations. Globally, Parker et al. (2016) reported that, in those countries with higher levels of ability stratification through academic selection, the expectations of students from lower socioeconomic groups in relation to higher education participation were more strongly correlated with students'

early academic achievement than in countries with less ability stratification. Therefore, it is unsurprising that academic selection strengthens the correlation between socioeconomic background and students' propensity to progress to higher education. In a similar vein, the greater inequalities in adult earnings of individuals who grew up in selective rather than comprehensive education systems resonate with Borghans et al.'s (2020) contention that early academic selection tends to have a detrimental effect on the future earnings of low- and medium-ability students.

The findings from the current review indicate that academic selection does little to promote equity in educational outcomes for students from different socioeconomic backgrounds but, rather, it reinforces and further exacerbates socioeconomic disparities in educational outcomes. This also aligns with the international evidence pertaining to the way academic selection serves to perpetuate and entrench socioeconomic inequity in educational outcomes (Bol & Van de Werfhorst, 2013; Horn, 2009; Terrin & Triventi, 2022; Van de Werfhorst & Mijs, 2010). For example, Bol and Van de Werfhorst (2013) found that, in education systems with higher levels of ability stratification, students' academic achievement levels were more strongly correlated with their socioeconomic backgrounds than in less stratified systems. Horn (2009) found a similar pattern: that academic selection exacerbated attainment inequalities by socioeconomic background, and that an earlier selection age was associated with greater socioeconomic background attainment differentials. Contrastingly, Horn (2009) concluded that comprehensive post-primary education systems were associated with lower socioeconomic background attainment differences. These findings are further reinforced by Terrin and Triventi's (2022) conclusion, from their meta-analysis of the effects of ability grouping on academic achievement, that higher levels of ability stratification increased the correlation between students' social origins and their attainment levels.

The results of the evidence synthesis indicated that the transition from a selective to a comprehensive education system in England had no statistically significant impact on inter-generational social mobility. In other words, an academically selective system was no better at promoting social mobility than the comprehensive system that replaced it in most areas of England. The failure of selective systems to promote social mobility aligns with Hurn's (1993) thesis that, even if supposedly meritocratic systems that condone academic selection result in students from less privileged backgrounds gaining better qualifications than they would have obtained in a comprehensive system, their more affluent peers can restore their competitive advantage in the pursuit of future employment opportunities through their access to greater financial and other resources. Consequently, the type of education system within which students are educated has limited influence on their longer-term career prospects, and the potential for social mobility is therefore correspondingly compromised.

Limitations

The main strength of the current review is that it investigates the consequences of academic selection relative to comprehensive education at post-primary level in a context that operates both systems, that is, the United Kingdom. This permitted the evidence pertaining to the relative advantages and disadvantages of the two systems to be critically evaluated. However, the review also had some limitations. Due to time and resource restrictions, only publications written in English and relating to the UK context were included in the review. Given that academic selection, or between-school tracking, does exist in some other countries, such as Germany and Hungary, for example, this is a limitation of the review. Although the inclusion of publications in languages other than English and/or about contexts other than the United Kingdom may have impacted upon the conclusions of the review, comparatively few countries retain a narrowly focused system of early academic selection system

like that in those areas of the United Kingdom that condone selection. Therefore, evidence pertaining to academic selection consequences in other jurisdictions may not have been directly comparable to the UK evidence.

IMPLICATIONS FOR POLICY, PRACTICE AND FUTURE RESEARCH

The evidence reviewed suggests that there are benefits of a grammar school education for those who are fortunate enough to obtain one, particularly in relation to improved academic achievement and progression to higher education. However, it is also evident that there are some difficulties associated with academic selection in the UK context. The dominant issues of concern are the failure of academic selection to promote social mobility and the way in which it appears to reinforce and further entrench socioeconomic disparities in educational outcomes. The failure to promote social mobility is a grave concern given that the promotion of intergenerational social mobility is the main *raison d'être* of academic selection. Academic selection's apparent exacerbation of socioeconomic disparities in educational outcomes is problematic since it effectively leads to the Matthew effect, whereby more socially privileged individuals accrue further advantage with time, while the less privileged are further disadvantaged over time. This is problematic since it implies that academic selection precipitates cultural reproduction, whereby social inequalities and class structures are perpetuated across generations. Interestingly, scant evidence emerged from the review to support technical-functional theories of academic selection, which argue that selection serves a necessary function by ensuring that students receive an education suited to their skills and abilities, and which prepares them efficiently for their roles in society (Kelly & Price, 2011). Therefore, we suggest that, in the short term, appropriate interventions should be implemented in non-grammar schools within selective areas to address underachievement. However, in the longer term, the future of academic selection needs to be reviewed, and options for replacing it with more socially just approaches to educating children and young people should be explored.

The review also revealed some gaps in the current evidence base that warrant further research. More specifically, the review revealed a dearth of studies that explore the impact of academic selection on students' longer-term health, happiness, personal autonomy and civic responsibility, which we contend are at least as important educational outcomes as academic achievement or financial rewards. Pastore and Jones (2023) was the only included study which considered the longer-term effect of grammar school attendance on any of these outcomes, concluding that there was no significant difference in health outcomes between those who attended grammar schools and those who did not. Therefore, the longer-term impact of academic selection on such things as health, happiness, personal autonomy and civic responsibility is an important area for future research.

CONCLUSION

In this review, our objective was to critically evaluate the quantitative research evidence pertaining to the consequences of academic selection for post-primary education in the United Kingdom.

The main findings from the review include:

1. When relevant student and school characteristics are controlled for, academic selection has a negligible overall effect, at the system level, on academic achievement relative to all-ability comprehensive education. Most of the reviewed evidence suggests

that grammar school attendance leads to small/moderate positive effects on student achievement relative to comparable students who attend comprehensive schools, but this is counterbalanced by small negative achievement effects for students in non-grammar schools within selective areas relative to comparable students who attend comprehensive schools. Grammar school attendance increases the likelihood of students' participation in higher education relative to comparable students who attend other types of schools. Academic selection has limited impact on longer-term labour market and earnings-related outcomes. There are minimal differences in socio-emotional outcomes, such as well-being and self-esteem, between grammar and comparable non-grammar students.

2. Academic selection serves to reinforce and further exacerbate socioeconomic disparities in educational outcomes.
3. There is minimal difference in intergenerational social mobility between selective and comprehensive education systems.

Although the number of articles included in the current systematic review was relatively small ($n=32$), and the included studies are of variable quality, the work has important implications for both future research and policy and practice. There is a dearth of evidence pertaining to the longer-term impact of academic selection on such things as health, happiness, personal autonomy and civic responsibility, and these are therefore important areas for future research. Given that there is limited evidence to suggest that academic selection promotes social mobility, despite this being an important rationale for its use, coupled with its deleterious consequences for many children, the future of academic selection needs to be reviewed, and options for replacing it with more socially just approaches to post-primary education should be explored.

AUTHOR CONTRIBUTIONS

Ian Cantley: Conceptualization; methodology; investigation; writing – review and editing; funding acquisition; writing – original draft; formal analysis; validation. **Rebecca Loader:** Conceptualization; methodology; investigation; writing – review and editing; funding acquisition; validation. **Joanne Hughes:** Conceptualization; methodology; investigation; funding acquisition; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors report there are no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ETHICS STATEMENT

This systematic review did not involve the collection of primary data from human participants; therefore, ethical approval and informed consent were not required. The review was conducted with a commitment to transparency, objectivity, and academic integrity, and all efforts were made to ensure the accuracy and reliability of the data included.

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APPENDIX 1

PRISMA 2020 checklist

Section and topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review	p. 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	p. 1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge	p. 7
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses	p. 9
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	p. 11
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	p. 10
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used	p. 10
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process	p. 13
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process	p. 13
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect	N/A
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	N/A

APPENDIX 1 (Continued)

Section and topic	Item #	Checklist item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process	p. 14
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	p. 13
	13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	p. 14
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression)	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	p. 14
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	p. 14

RESULTS

Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Figure 1
Study characteristics	17	Cite each included study and present its characteristics	Appendix 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study	Appendix 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g., confidence/credible interval), ideally using structured tables or plots	N/A

APPENDIX 1 (Continued)

Section and topic	Item #	Checklist item	Location where item is reported
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	p. 15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	Appendix 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence	p. 25
	23b	Discuss any limitations of the evidence included in the review	p. 28
	23c	Discuss any limitations of the review processes used	p. 28
	23d	Discuss implications of the results for practice, policy, and future research	p. 29
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	N/A
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	p. 1
Competing interests	26	Declare any competing interests of review authors	p. 3
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	N/A

Note: From Page et al. (2021).

APPENDIX 2

Results of included studies

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
1	Ahmavaara and Houston	2007	N=856 53.5% from 2 grammar schools, 46.5% from 2 non-selective secondary modern schools in England; 55.3% 11–12 years old, 44.7% 14–15 years old; 50.1% female, 49.9% male	Socio-emotional outcomes	Regression analysis	Grammar school attendance positively predicted confidence in own intelligence ($\beta=-0.08$, $p<0.05$), perceived academic performance ($\beta=0.273$, $p<0.001$), and future achievement aspirations ($\beta=0.529$, $p<0.001$), but did not predict self-esteem ($\beta=-0.031$, $p>0.05$) or identification with school ($\beta=-0.063$, $p>0.05$). Grammar school attendance also indirectly influenced future achievement aspirations, with effects mediated by perceived academic performance and confidence in own intelligence. Students' views about the nature of intelligence (whether it is fixed or malleable) had a stronger influence on future achievement aspirations for grammar school students than for non-grammar school students. Non-selective students' views about intelligence had no significant effect on their future achievement aspirations, but grammar school students with more fluid views of intelligence had significantly higher future achievement aspirations ($\beta=-0.243$, $p<0.001$).	Very weak design based on cross-sectional study of students from 4 schools, with no attempt to compare grammar and non-selective students with similar baseline characteristics. Used OLS regression rather than multi-level modelling, which would have accounted for hierarchical nature of the data	1
2	Andrews, Hutchinson, and Johns	2016	Total number of cases not reported, but it was large since it was based on 2014 and 2015 GCSE results for students in England. However, propensity score-matched sample size that was used to assess impact of academic selection on achievement was not reported	Academic achievement and socioeconomic equity in educational outcomes	Summary statistics for propensity score-matched samples	When student characteristics were controlled for, academic selection led to no overall effect, either positive or negative, on academic achievement at GCSE level. On average, in selective areas, grammar school students achieved 2.3 GCSE grade points higher than similar students from non-selective schools in comprehensive areas did, but students who did not attend grammar schools achieved 0.6 of a grade point lower than comparable students from non-selective schools in comprehensive areas did. However, in the most selective areas with the highest proportions of available grammar school places, the positive grammar school effect on achievement was lower at just 0.8 of a grade point. The positive grammar school effect was higher for free school meal (FSM) students (approximately 3.4 GCSE grade points) than non-FSM students, but the penalty of not attending a grammar school was also greater for FSM students at approximately 1.2 grade points. The difference in the percentage of students obtaining five GCSE A*-C grades between FSM eligible and non-FSM eligible students was greater in areas operating a fully selective system (approximately 34.1%) than in non-selective (comprehensive) areas (approximately 27.8%)	Non reporting of propensity score-matched sample size and comparison group sizes	2

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
3	Atkinson, Gregg, and McConnell	2006	N=491,436 Analysis based on student level data from national databases in England for mainstream state secondary school students in 2002 with known GCSE results	Academic achievement	Regression analysis with propensity score-matching	Overall, academic selection had limited impact on academic achievement. Grammar school students achieved significantly higher GCSE results (equivalent to approximately 4 grade points) than students with comparable KS2 attainment did during the latter stages of primary education in similar non-selective regions. However, non-grammar school students within selective regions were disadvantaged by just under one grade point compared to similar students in non-selective regions	Grammar school effects were estimated using KS2 results from primary education rather than selection test results. Used OLS regression rather than multi-level modelling	3
4	Birkelund, Capsada-Munsech, and Karlson	2021	N=9655 individuals from the longitudinal 1958 National Child Development Study in Great Britain	Earnings	Regression analysis	Grammar school attendance led to a modest cumulative lifetime earnings return of just £39,000 relative to attendance at a non-grammar state school. However, 85% of this was explained by socioeconomic background and the ability levels of the individuals rather than school type attended	Weak design that did not adequately deal with non-random allocation of individuals to grammar schools. Earnings at most time points were estimated using linear interpolation of values between adjacent time points for which data were collected (at ages 23, 33, 42, 46, and 55). Lower ability and lower socioeconomic background individuals were under-represented in the sample. The study was based on analysis of data pertaining to individuals who were born in 1958 and so findings may not be generalisable to the modern era	2
5	Boliver and Swift	2011	N=3335 for income mobility, N=4728 for social class mobility (from the longitudinal 1958 National Child Development Study in Great Britain in both cases)	Income and social class mobility	Binary logistic regression analysis with propensity score-matching	Selective education offered no overall statistically significant income or social class mobility advantage relative to comprehensive education for individuals from any origin. The negative effects associated with attending a secondary modern school offset any positive grammar school effect for low origin individuals	Some of the comparison groups had relatively small numbers of cases. The study was based on analysis of data pertaining to individuals who were born in 1958 and so findings may not be generalisable to the modern era, particularly since the comprehensive system was in its infancy when the individuals were in post-primary education	3

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
6	Burgess, Crawford, and Macmillan	2017	Large samples based on linked data from National Pupil Database and Higher Education Statistics Agency for students from England who sat their GCSEs between 2003 and 2006 and potentially entered higher education between 2005–2006 and 2009–2010	Higher education participation and outcomes	Regression analysis based on matched samples	Grammar school students were 22% more likely to progress to higher education and 17% more likely to attend a prestigious university than students who just missed out on securing a grammar school place in selective regions. After controlling for prior attainment, grammar school students were less likely to satisfactorily complete their courses or to graduate with good degree classifications than students who just missed out on securing a grammar school place in selective regions. Students who marginally failed to secure a grammar school place in selective regions were 3% less likely to pursue higher education, 8% less likely to attend a prestigious institution, and 3% less likely to achieve a good degree classification than comparable students from non-selective regions were	Matching processes employed may have compromised the robustness of the findings	3
7	Burgess, Dickson, and Macmillan	2020	N=2033 Data derived from the large, longitudinal Understanding Society panel study that has followed over 40,000 UK households from 2009	Earnings and socioeconomic equity in educational outcomes	Regression analysis based on matched samples	The adult earnings of individuals who grew up in a selective education system showed significantly greater inequality than for those who grew up under a comprehensive system, with large effect sizes: 24% of the raw 90–10 earnings differential and 19% of the conditional 90–10 earnings differential were attributable to the differences between selective and comprehensive education systems	The matching of selective and non-selective areas based on several characteristics that could influence that future earnings did not guarantee that the individuals in the selective and non-selective areas were appropriately matched on baseline attributes that could influence future earnings. Greater diversity of individual baseline characteristics in selective regions may have led to the observed earnings differential rather than the selective education system	3
8	Buscha, Gorman, and Sturgis	2023	N=90,894 Based on longitudinal census data linked to administrative data on school selectivity in England and Wales	Social mobility	Regression analysis with matched samples	The transition from a selective to a comprehensive education system in England had no statistically significant impact on intergenerational social mobility	Type of school attended by an individual was inferred from the predominant type of education system in the area where they lived at the time, which may have introduced measurement errors	3
9	Capsada-Munsech and Boliver	2024	N=5400 71.1% grammar students, 92.9% non-selective (comprehensive) students from Next Steps longitudinal survey of individuals born in England during 1989–1990 and linked data from the National Pupil Database	Higher education participation	Logistic regression analysis	Grammar school students were no more likely to participate in higher education or to attend prestigious Russell group universities than students from non-selective (comprehensive) state schools with comparable GCSE and A-level attainment, and irrespective of socioeconomic background	Controlling for students' A-level attainment may be over-controlling given the strong correlation between A-level attainment and higher education participation. Relatively small number of grammar students included in sample and limited reporting of information on missing data	2

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
10	Clark	2010	Sample size varied for each aspect of the study but was at least $N=1155$. Based on a longitudinal study of individuals from England who left school in the early 1970s	Academic achievement and higher education participation	Regression discontinuity design	After controlling for performance in the selection test, gender and socioeconomic status, 4 years of grammar school attendance had a non-significant effect on academic achievement scores, with a small effect size of around one sixth of a standard deviation in favour of grammar school students. Grammar school students were more likely than non-selective school students were to progress to university	Although the study used a strong regression discontinuity design that permitted estimation of causal effects, the findings may not be generalisable to the current era given the changes to the education system since the 1960s and 1970s. For example, modern approaches to academic selection are very different those used in the 1960s. The other changes to the education system since the 1960s and 1970s, such as differences in student characteristics, curricula, and academic standards, are also likely to compromise the findings of this study	3
11	Coe, Jones, Searle, Kokotsaki, Kosnin, and Skinner	2008	Various analyses were performed but, for the most robust analysis, $N=28,200$ 50% grammar students, 50% non-grammar students based on propensity-matched samples of students from National Pupil Database in England	Academic achievement and socioeconomic equity in educational outcomes	Regression analysis with propensity score-matched samples	Grammar school students perform marginally better at GCSE than comparable non-grammar students, with a differential of between 0 and 0.75 of a GCSE grade per subject taken. The positive grammar effect decreased as more factors were controlled for in the regression analysis, so the true grammar effect is likely to be at the lower end of the 0 to 0.75 GCSE grade per subject range. Overall, FSM eligible students obtained just under 0.5 of a GCSE grade less than comparable non-FSM eligible students did, but the FSM-eligible students were marginally less disadvantaged relative to their comparable non-FSM eligible peers if they attended a grammar school, with the difference equating to approximately 0.125 of a GCSE grade. Although grammar students made greater progress than non-grammar students during their grammar school careers, they also made greater progress during their primary school careers, thus suggesting there may have been unaccounted for differences between the two groups of students and consequently reducing confidence in a positive grammar school effect	Propensity score matching was performed using KS2 test results from primary education rather than selection test results	3
12	Crawford	2014	$N=4,363,600$ Based on linked individual-level data for GCSE students between 2001–2002 and 2007–2008 from various national databases in England	Higher education participation and outcomes	Logistic regression analysis	State grammar school students were around 4% more likely to participate in higher education and around 1% more likely to enrol in a prestigious university than comparable non-selective state school students were. Grammar students were significantly less likely to finish their courses and to attain a good degree classification than comparable non-selective students were	Weak design that did not permit causal relationships to be established (e.g., no use of regression discontinuity design or propensity score matching)	2

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
13	Early	2020	<i>N</i> = 61,373 Based on linked individual-level data for 3 consecutive cohorts of students in Northern Ireland who completed GCSEs in 2010–2011, 2011–2012, and 2012–2013	Academic achievement and socioeconomic equity in educational outcomes	Multi-level regression analysis	After controlling for a range of factors, grammar school students obtained 24.13 GCSE grade points more than non-grammar students did. FSM eligible grammar students achieved significantly higher GCSE results than FSM eligible non-grammar students did ($d = 1.45$). FSM eligible grammar students achieved significantly higher GCSE results than non-FSM eligible non-grammar students did ($d = 1.23$), and there was no significant difference between the GCSE performance of FSM eligible and non-FSM eligible grammar students ($d = 0.02$).	Very weak design that did not permit causal relationships to be established (e.g., no use of regression discontinuity design or propensity score matching). Students' prior attainment was not controlled for	1
14	Galindo-Rueda and Vignoles	2005	<i>N</i> = 7913 40.4% selective system (grammar or secondary modern school), 59.6% comprehensive system from the longitudinal 1958 National Child Development Study in Great Britain	Academic achievement	Regression analysis with propensity score-matching	There was no systematic statistically significant difference in educational achievement between selective and comprehensive education systems. The highest ability students, particularly girls, achieved significantly higher academic standards in selective systems than similar ability students in comprehensive systems. There was no systematic difference between the academic achievement of lower ability students in selective and comprehensive systems	Local selection conditions were inferred from the school attended which may have introduced measurement errors. The study was based on analysis of data pertaining to individuals who were born in 1958 and so findings may not be generalisable to the modern era, given the changes to the education system since the 1960s and 1970s	2
15	Gallagher and McKeown	2000	<i>N</i> = 2130 44.6% grammar school students, 55.4% secondary (non-selective) school students from Northern Ireland	Socio-emotional outcomes	Analysis of variance	There was no statistically significant difference between grammar and secondary students' overall satisfaction (positive affect) levels with school, but grammar students' negative affect levels were significantly greater than those for secondary students ($F = 6.8$, $p < 0.01$), thus indicating a more negative disposition to their schools than secondary students. However, both grammar and secondary negative affect levels were small in magnitude in comparison to the corresponding positive affect levels, thus suggesting that students from both sectors were positively disposed towards their schools overall. Secondary students had a statistically significantly more positive perception of the relationship between teachers and students in their schools than their grammar counterparts ($F = 23.3$, $p < 0.001$), secondary students perceived their schooling to be more relevant for their future than grammar students ($F = 5.8$, $p < 0.05$), grammar students had greater confidence in their ability to succeed in school work than their secondary counterparts ($F = 5.9$, $p < 0.05$), there was no statistically significant difference between grammar and secondary students in relation to their perceptions of the status accorded to them by others within their schools, secondary students had significantly higher perceived levels of social integration than those from grammar schools ($F = 5.0$, $p < 0.05$)	Very weak design that did not permit causal inferences to be established	1

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
16	Gorard and Siddiqui	2018	N=549,203 4.1% grammar students, 95.9% non-grammar students from the 2015 cohort of GCSE students in England included in the National Pupil Database	Academic achievement and socioeconomic equity in educational outcomes	Regression analysis	After controlling for a range of background variables, including the number of years for which students had been eligible for free school meals, grammar school students' achievement levels were not appreciably higher than anticipated ($\beta=0.04$). The poverty attainment gap (difference between attainment of FSM and non-FSM eligible students) was lower in magnitude for grammar students (effect size= -0.41) than the general population (effect size= -0.69). However, the poverty attainment gap for students from selective areas had an effect size of -0.76 , suggesting that the benefits of grammar school attendance for FSM students were negated by the harmful effects experienced by students who did not secure grammar school places in selective areas	Weak design that did not permit causal inferences to be established. Relatively small number of grammar school students included in analysis	2
17	Harris and Rose	2013	N=934 (Buckinghamshire grammar & secondary modern) N=1582 (Buckinghamshire grammar & Oxfordshire comprehensive) N=2040 (Buckinghamshire secondary modern & Oxfordshire comprehensive) 50% matched samples from each of the two groups represented in each of the three samples (based on prior attainment and other relevant variables)	Academic achievement	Logistic regression analysis with matched samples	In Buckinghamshire (a selective area), grammar students were 15.2% more likely than comparable secondary modern students to obtain at least five GCSE grades A*-C inclusive of English and mathematics. Grammar students from Buckinghamshire were 8.4% more likely than comparable comprehensive students from Oxfordshire were to obtain at least five GCSE grades A*-C inclusive of English and mathematics. Secondary modern students from Buckinghamshire were 6.4% less likely than comparable comprehensive students from Oxfordshire were to obtain at least five GCSE grades A*-C inclusive of English and mathematics	Although a strong design was used, matching was based on KS2 results from primary education rather than selection test results	3
18	Jerrim and Sims	2019a	N=4785 Derived from the longitudinal Millennium Cohort Study of children born in the UK in 2000; 22.9% from selective area, 77.1% from comprehensive area of England	Socio-emotional outcomes	Regression analysis and propensity score-matching	There were minimal differences in socio-emotional outcomes between students from selective areas and comparable students from comprehensive areas. Within selective areas, grammar school students had greater educational aspirations than their non-grammar counterparts, e.g., effect size for 14-year-old students expecting to remain in school post-16= 0.320 , effect size for 14-year-old students expecting to progress to university= 0.301	Although the study used a fairly strong design, there may have been unobserved differences between the comparison groups that were not adequately accounted for without using an even stronger design, e.g., regression discontinuity design	3

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
19	Jerrim and Sims	2020	Slight variations in sample size, but most analyses based on approximately $N=1150$. Propensity-matched sample from the longitudinal Millennium Cohort Study of children born in the UK in 2000; approximately 56.5% from England and 43.5% from Northern Ireland	Socio-emotional outcomes	Regression analysis and propensity score-matching	In both England and Northern Ireland, there were minimal differences between grammar school students and their comparable non-grammar peers in most of the socio-emotional outcomes considered (e.g., well-being, self-esteem and future aspirations). Northern Irish grammar students reported significantly lower academic self-concept levels than their comparable non-grammar peers (effect size = -0.38)	Although a fairly strong design was used, findings may have been compromised by the relatively small sample size and the small number of cases in some sub-groups, e.g., grammar school students from lower socioeconomic backgrounds	2
20	Levačić and Marsh	2007	$N=340,124$ Based on national dataset for students in England who sat GCSEs in 2001	Academic achievement	Multi-level regression analysis	Grammar students obtained 5.5 GCSE grade points more than comparable students who attended comprehensive schools, and secondary modern students achieved one GCSE grade point less than comparable comprehensive students did	Weak design that did not permit causal inferences to be established	2
21	Lu	2020a	The findings in this publication are replicated in some of the other included publications involving this author but, in an additional relevant study within this report, $N=149,072$. Based on valid records from the National Pupil Database for students from the 36 selective local authorities in England (i.e., those with any grammar schools) who sat GCSEs in 2015–2016	Academic achievement and socioeconomic equity in educational outcomes	Regression analysis	FSM-eligible students who attended grammar schools achieved approximately 1.2 GCSE grade points more than comparable FSM-eligible students who attended non-selective schools within the 36 local authorities included in the analysis	Weak design that did not permit causal inferences to be established	2
22	Lu	2020b	$N=2541$ from one local authority in England with a high proportion of grammar school places who sat the grammar school selection test in 2011	Academic achievement	Regression discontinuity design	Grammar students obtained approximately 4.5 GCSE grade points more than comparable non-grammar students	Although a strong regression discontinuity design was used, the focus on one geographical area allied with missing/incomplete data issues may have impacted on the findings	3
23	Lu	2021	$N=438,000$ Based on linked data from National Pupil Database and Higher Education Statistics Agency for a cohort of students from England who commenced higher education in 2015–2016	Higher education participation	Logistic regression analysis	Grammar school students were 24% more likely to pursue higher education than comparable comprehensive school students from non-selective areas with similar academic attainment up to GCSE level were. Grammar school students were 19% more likely to progress to a prestigious Russell group university than comparable comprehensive school students from non-selective areas with similar academic attainment up to GCSE level were	Weak design that did not permit causal inferences to be established	2

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
24	Lu, Anders, Siddiqui, and Shao	2024	<i>N</i> = 481,681 Based on the cohort of students from the National Pupil Database in England who sat GCSE examinations during 2015–2016	Academic achievement	Regression analysis	There was no significant overall difference in academic achievement between academically selective and non-selective (comprehensive) systems after students' baseline characteristics were controlled for. Academic selection disadvantaged high-attainers since students from selective areas were less likely than comparable students from non-selective (comprehensive) areas were to obtain at least five GCSE A*–A grades	Weak design that did not permit causal inferences to be established	2
25	Lu, Dai, See, Shao, and Hu	2024	<i>N</i> = 438,000 Based on linked data from National Pupil Database and Higher Education Statistics Agency for a cohort of students from England who commenced higher education in 2015–2016	Higher education participation and socioeconomic equity in educational outcomes	Logistic regression analysis	Academic selection strengthened the correlation between students' early and later academic attainment and the relationship between family background and higher education participation	Weak design did not permit causal inferences to be established and insufficient evidence was presented to warrant the claim about the relationship between family background and higher education participation (as early academic attainment was taken as a proxy for family background)	1
26	Lu and Siddiqui	2024	<i>N</i> = 149,072 Based on valid records from the National Pupil Database for students from the 36 selective local authorities in England (i.e., those with any grammar schools) who sat GCSEs in 2015–2016	Academic achievement	Regression analysis	Overall, grammar school students achieved 0.3 GCSE grade points more than comparable students from non-selective schools within the same 36 local authorities were. There was no systematic relationship between the capacity of grammar schools to improve academic achievement relative to non-selective schools within the same local authority and the proportion of grammar school places within the authority, but local authorities with consistently positive grammar school effects had smaller proportions of grammar school places available	Weak design that did not permit causal inferences to be established	2
27	Mansfeld	2019	<i>N</i> = 182,875 Based data from Department for Education in England for student destinations after KS5 in 2016	Higher education participation and social mobility	Descriptive statistics and basic hypothesis tests	Students from selective areas were 1.1 times more likely to progress to higher education, 1.7 times more likely to progress to highly selective higher education, and 2.6 times more likely to go to Oxbridge than students from comprehensive areas. Students from the most disadvantaged backgrounds were more than twice as likely to go to Oxbridge if they resided in a selective rather than a non-selective region. Grammar schools exert a considerable influence on social mobility	Limited attempt to control for relevant factors that may have influenced the outcomes, e.g., using propensity score matching and/or regression analysis. No consideration of the outcomes for students who did not gain a grammar school place or did not progress to higher education. Insufficient evidence presented to validate the claim that grammar schools exert a considerable influence on social mobility	0

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
28	Montacute and Cullinane	2018	Large number of cases based on higher education application data from schools and colleges in England for 3 consecutive cycles (2014–2015, 2015–2016, and 2016–2017)	Higher education participation	Descriptive statistics	63% of applicants from grammar schools received and accepted offers from prestigious Russell group universities, compared to 44% of applicants from non-selective state schools. Oxbridge acceptance rates were higher for grammar school applicants (31%) than applicants from other state schools (25% acceptance rate for sixth form college applicants and 22% acceptance rate for comprehensive school applicants)	Limited attempt to control for relevant factors that may have influenced the outcomes, e.g., using propensity score matching and/or regression analysis	0
29	Pastore and Jones	2023	N varied between 1450 and 2800 individuals from the longitudinal 1958 National Child Development Study in Great Britain	Academic achievement, labour market outcomes, health outcomes	Regression discontinuity design	Grammar school attendance positively predicted academic achievement but had no significant impact on longer-term labour market or health outcomes	Although based on a strong regression discontinuity design, the study was based on analysis of data pertaining to individuals who were born in 1958 and so findings may not be generalisable to the modern era. Transition from selective to comprehensive education in England during the period when the individuals would have been in post-primary education may have impacted on the findings	3
30	Schagen and Schagen	2003	N=387,595 (KS2 [age 11 years] to KS3 [age 14 years] analysis) N=482,399 (KS3 to GCSE analysis) Based on national value-added datasets in England for KS2 1997 to KS3 2000 and KS3 1998 to GCSE 2000 results	Academic achievement	Regression analysis	Grammar students performed significantly better at KS3 than comparable students who did not attend grammar schools. Overall, grammar students performed significantly better at GCSE (by approximately three grade points) than comparable students who did not attend grammar schools, but the grammar school advantage disappeared for students with the highest levels of prior attainment	Weak design that did not permit causal inferences to be established	2
31	Shuttleworth and Daly	2000	N=1784 Based on approximately equal numbers of grammar and non-grammar students from 8 grammar and 17 secondary schools in Northern Ireland, who sat GCSEs in the 1998–1999 academic year	Academic achievement	Multi-level regression analysis	Grammar school students significantly outperformed their comparable non-grammar school peers by approximately 16 GCSE grade points	Weak design that did not permit causal inferences to be established	2

APPENDIX 2 (Continued)

Number	Author(s)	Year	Number of individuals and data source	Main outcome(s) investigated	Analytical approach	Summary of main findings	Main limitations	Rating
32	Sullivan, Parsons, Green, Wiggins, Ploubidis, and Huynh	2018	N=10,188 from the longitudinal 1970 British Cohort Study in England and Wales	Academic achievement (short-term and longer-term)	Regression analysis	Grammar students achieved significantly better academic qualifications than comparable comprehensive students did both in the short term (ages 16 and 18) and the longer term (age 42). Secondary modern students achieved significantly lower academic qualifications than comparable comprehensive students did at age 16 and the differential persisted in relation to the highest qualification level achieved by age 42	Weak design that did not permit causal inferences to be established. Students' prior attainment was not controlled for	2